Python Codes for Practice

Exercise 1:

Write a Python function that accepts two integer numbers. If the product of the two numbers is less than or equal to 1000, return their product; otherwise, return their sum.

Given Input:

- Case 1: number1 = 20, number2 = 30
- Case 2: number1 = 40, number2 = 30

Expected Output:

- The result is 600
- The result is 70

Exercise 2:

Iterate through the first 10 numbers (0–9). In each iteration, print the current number, the previous number, and their sum.

Given Input: Range: numbers = range(10)

Expected Output:

Printing current and previous number sum in a range(10)
Current Number 0 Previous Number 0 Sum: 0
Current Number 1 Previous Number 0 Sum: 1
Current Number 2 Previous Number 1 Sum: 3
....
Current Number 8 Previous Number 7 Sum: 15
Current Number 9 Previous Number 8 Sum: 17

Exercise 3:

Display only those characters which are present at an even index number in given string.

Given Input: String: "pynative"

Expected Output:

Original String is pynative
Printing only even index chars
p
n
t
v

Exercise 4:

Write a function to remove characters from a string starting from index 0 up to n and return a new string.

Given Input:

- remove_chars("pynative", 4)
- remove_chars("pynative", 2)

Expected Output:

- tive
- native

Exercise 5:

Write a program to swap the values of two variables, `a` and `b`, without using a third temporary variable.

Given Input: a = 5, b = 10

Expected Output:

Before Swap: a = 5, b = 10
After Swap: a = 10, b = 5

Exercise 6:

Write a program that calculates the factorial of a given number (e.g., 5!) using a `for` loop.

Given Input: number = 5

Expected Output: The factorial of 5 is 120

Exercise 7:

Create a list of 5 fruits. Add a new fruit to the end of the list, then remove the second fruit (at index 1).

Given Input: fruits = ["apple", "banana", "cherry", "date", "elderberry"]

Expected Output: ['apple', 'cherry', 'date', 'elderberry', 'fig']

Exercise 8:

Write a program that takes a string and reverses it (e.g., “Python” becomes “nohtyP”).

Given Input: text = "Python"

Expected Output: Reversed: nohtyP

Exercise 9:

Write a program to count the total number of vowels (a, e, i, o, u) present in a given sentence.

Given Input: sentence = "Learning Python is fun!"

Expected Output: Number of vowels: 6

Exercise 10:

Given a list of integers, find and print both the largest and the smallest numbers.

Given Input: nums = [45, 2, 89, 12, 7]

Expected Output: Largest: 89 Smallest: 2

Exercise 11:

Write a script that takes a list containing duplicate items and returns a new list with only unique elements.

Given Input: data = [1, 2, 2, 3, 4, 4, 4, 5]

Expected Output: Unique List: [1, 2, 3, 4, 5]

Exercise 12:

Write a function to return `True` if the first and last number of a given list is the same. If the numbers are different, return `False`.

Given Input:

- numbers_x = [10, 20, 30, 40, 10]
- numbers_y = [75, 65, 35, 75, 30]

Expected Output:

Given list: [75, 65, 35, 75, 30] | result is False
Given list: [10, 20, 30, 40, 10] | result is True

Exercise 13:

Iterate through a given list of numbers and print only those numbers which are divisible by 5.

Given Input: num_list = [10, 20, 33, 46, 55]

Expected Output:

Divisible by 5:
10, 20, 55

Exercise 14:

Write a program to find how many times the substring “Emma” appears in a given string.

Given Input:

- str_x = "Emma is good developer. Emma is a writer"

Expected Output: Emma appeared 2 times

Exercise 15:

Print the following pattern where each row contains a number repeated a specific number of times based on its value.

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

Given Input: Range: 1 to 5

Expected Output: (The pattern shown above)

Exercise 16:

Write a program to check if a given number is a palindrome (reads the same forwards and backwards).

Given Input:

- Case 1: number = 121
- Case 2: number = 125

Expected Output:

Number 125 is not palindrome number
Number 121 is palindrome number

Exercise 17:

Create a new list from two given lists such that the new list contains odd numbers from the first list and even numbers from the second list.

Given Input:

- list1 = [10, 20, 25, 30, 35]
- list2 = [40, 45, 60, 75, 90]

Expected Output: [25, 35, 40, 60, 90]

Exercise 18:

Write a program to extract each digit from an integer in the reverse order.

Given Input: number = 7536

Expected Output: 6 3 5 7

Exercise 19:

Calculate income tax for a given income based on these rules:

- First $10,000: 0% tax
- Next $10,000: 10% tax
- Remaining income: 20% tax

- **Given Input:** `income = 45000`
- **Expected Output:** Total income tax to pay is 6000

Exercise 20:

Print a multiplication table from 1 to 10 in a formatted grid.

Given Input: Range: 1 to 10

Expected Output:

1 2 3 4 5 6 7 8 9 10
2 4 6 8 10 12 14 16 18 20
... (up to 10)

Exercise 21:

Print a downward half-pyramid pattern using stars (*).

Given Input: Rows: 5

Expected Output:

```
* * * * *
* * * *
* * *
* *
*
```

Exercise 22:

Write a function called `exponent(base, exp)` that returns an integer value of the base raised to the power of the exponent.

Given Input: base = 2, exp = 5

Expected Output: 2 raises to the power of 5: 32

Exercise 23:

Write a program to check if a given number is a palindrome. A palindrome number remains the same when its digits are reversed (e.g., 121, 545).

Given Input: number = 121

Expected Output:

Original number 121

Yes. given number is palindrome number

Exercise 24:

Write a program to print the first 15 terms of the Fibonacci series. The sequence starts with 0 and 1, and each subsequent number is the sum of the two preceding ones.

Given Input: Terms = 15

Expected Output: 0 1 1 2 3 5 8 13 21 34 55 89 144 233 377

Exercise 25:

Write a program that takes a year as input and determines if it is a leap year.

A **leap year** is a year in the Gregorian calendar that contains an extra day, making it 366 days long instead of the usual 365. This extra day, **February 29th**, is added to keep the calendar synchronized with the Earth's revolution around the Sun.

Rules for leap years: a year is a leap year if it's divisible by 4, unless it's also divisible by 100 but not by 400.

Given Input: year = 2024

Expected Output: 2024 is a leap year

Exercise 26:

Write a program that takes two separate dictionaries and merges them into one single dictionary.

Given Input:

dict1 = {"name": "Alice", "age": 25}

dict2 = {"city": "New York", "job": "Engineer"}

Expected Output:

{'name': 'Alice', 'age': 25, 'city': 'New York', 'job': 'Engineer'}

Exercise 27:

Take two lists and find the elements that appear in both. Use Sets to perform the operation.

Given Input:

list_a = [1, 2, 3, 4, 5]

list_b = [4, 5, 6, 7, 8]

Expected Output: Common Elements: {4, 5}

Exercise 28:

Start with a list of 10 numbers. Iterate through them and sort them into two separate lists: one for even numbers and one for odd numbers.

Given Input: numbers = [12, 7, 34, 21, 5, 10, 8, 3, 19, 2]

Expected Output:

Even numbers: [12, 34, 10, 8, 2]
Odd numbers: [7, 21, 5, 3, 19]

Exercise 29:

Create a list of 5 words. Write a loop that iterates through the list and prints each word alongside its character count.

Given Input: words = ["Apple", "Banana", "Cherry", "Date", "Elderberry"]

Expected Output:

Apple - 5 Banana - 6 Cherry - 6 Date - 4 Elderberry - 10

Exercise 30:

Write a program that counts how many times each word appears in a given paragraph and stores these counts in a dictionary.

Given Input: text = "apple banana apple cherry banana apple"

Expected Output: {'apple': 3, 'banana': 2, 'cherry': 1}

Exercise 31:

Write a program to find all prime numbers up to 20, but only print every second (alternate) prime number found.

Given Input: Limit = 20

Expected Output: 2, 5, 11, 17

Exercise 32:

Create a dictionary where the keys are numbers from 1 to 10 and the values are the squares of those numbers (e.g., 2: 4, 3: 9).

Given Input: Range: 1 to 10

Expected Output:

{1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64, 9: 81, 10: 100}

Exercise 33:

Ask the user for a sentence. Replace every empty space in that sentence with an underscore (_) and print the final result.

Given Input: "I love coding in Python"

Expected Output: I_love_coding_in_Python

Exercise 34:

Print a downward number pattern where each row starts with a decreasing value.

Given Input: Rows = 5

Expected Output:

5 4 3 2 1
4 3 2 1
3 2 1
2 1
1

Exercise 35:

Write a program to check if a user-entered string contains any numeric digits. Use a `for` loop to examine each character.

Given Input: input_string = "Python3"

Expected Output: The string 'Python3' contains digits: True

Exercise 36:

Write a program to capitalize the first letter of each word in a given string without using the built-in `.title()` method.

Given Input: text = "hello world from python"

Expected Output: Hello World From Python

Exercise 37:

Create a countdown timer that starts from a given number and counts down to zero using a `while` loop.

Given Input: start_count = 5

Expected Output: 5 4 3 2 1 Blast off!

Exercise 38:

Write a program that creates a new text file named `notes.txt`, writes three separate lines of text to it, and then reads that file back to display the contents in the console.

Given Input: Lines to write:

1. “Hello, this is my first note.”
2. “Python file handling is simple.”
3. “End of file.”

Expected Output:

notes.txt

Hello, this is my first note.
Python file handling is simple.
End of file.

Exercise 39:

Write a script that opens an existing `.txt` file and counts the total number of words it contains.

Given Input:

An external file sample.txt containing: *“Coding is the language of the future.”*

Expected Output: The file contains 7 words.

Exercise 40:

Create a `Car` class with attributes for `make`, `model`, and `year`. Include a method called `start_engine()` that prints a formatted string describing the car starting up.

Given Input: Make: “Toyota”, Model: “Camry”, Year: 2022

Expected Output: The 2022 Toyota Camry's engine is now running!